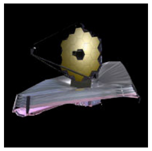

7937 - Cometary and Asteroidal Dust in our Zodiacal Cloud from Archival MIRI Observations of the 10-m Silicate Feature

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Most or all stars have orbiting debris, remnants of the process of planet formation. While the dust produced by colliding asteroids and sublimating comets is readily detected around many nearby stars, only in the solar system do we have a clear view of the overall system architecture: the spacing of the planets and the families of small bodies that produce such dust. Despite its proximity, the composition of the Solar System's zodiacal cloud of dust is not well known. Most is thought to come from Jupiter-family comets, but some models find a large fraction from asteroids. Dust from long-period comets (perhaps 10% of the total) has never been concludingly detected.

The best way to disentangle the various components of the zodiacal cloud is to look for differences with ecliptic latitude. Asteroids are closest to the midplane while Jupiter-family comets are more dispersed and long-period comets are isotropic.

Fortunately JWST provides a dataset ideal to address this issue. JWST regularly observes zodiacal dust, which manifests itself as a smooth mid-IR background. Normally considered noise to be removed, MIRI background observations are in fact scientific data waiting for analysis.

We propose to analyze the entire set of archived MIRI spectra spanning the entire sky, to measure the shape of the 10-m silicate feature. Webb's much improved sensitivity and stability compared with previous observations will enable much more sensitive tests for gradients in the zodiacal grains' composition and size distribution which would reveal the signatures of the different source populations in the only planetary debris disk where we can directly observe the parent bodies.

OBSERVING DESCRIPTION